

Overview of materials for Liquid Crystal Polymer (LCP), Unfilled

Categories: [Polymer](#); [Thermoplastic](#); [Liquid Crystal Polymer \(LCP\)](#); [Liquid Crystal Polymer \(LCP\), Unfilled](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Liquid Crystal Polymer (LCP), Unfilled". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: [Click here to view all available suppliers for this material.](#)

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Physical Properties	Metric		English	Comments
Density	1.38 - 2.02 g/cc		0.0499 - 0.0730 lb/in ³	Average value: 1.65 g/cc Grade Count:28
Water Absorption	0.0100 - 0.0400 %		0.0100 - 0.0400 %	Average value: 0.0217 % Grade Count:12
Moisture Absorption at Equilibrium	0.0200 - 0.0400 %		0.0200 - 0.0400 %	Average value: 0.0350 % Grade Count:4
Moisture Vapor Transmission	0.00606 - 0.354 cc-mm/m ² -24hr-atm	0.0154 - 0.900 cc-mil/100 in ² -24hr-atm		Average value: 0.0960 cc-mm/m ² -24hr-atm Grade Count:4
Oxygen Transmission	0.0154 - 0.0472 cc-mm/m ² -24hr-atm	0.0390 - 0.120 cc-mil/100 in ² -24hr-atm		Average value: 0.0296 cc-mm/m ² -24hr-atm Grade Count:4
Maximum Moisture Content	0.0100		0.0100	Average value: 0.0100 Grade Count:5
Linear Mold Shrinkage	0.000200 - 0.00400 cm/cm		0.000200 - 0.00400 in/in	Average value: 0.000960 cm/cm Grade Count:20
Linear Mold Shrinkage, Transverse	0.00280 - 0.00900 cm/cm		0.00280 - 0.00900 in/in	Average value: 0.00456 cm/cm Grade Count:19
Melt Flow	9.00 - 18.0 g/10 min		9.00 - 18.0 g/10 min	Average value: 13.7 g/10 min Grade Count:3

Mechanical Properties	Metric		English	Comments
Hardness, Rockwell M	45.0 - 90.0		45.0 - 90.0	Average value: 70.6 Grade Count:9
Tensile Strength, Ultimate	32.0 - 182 MPa		4640 - 26400 psi	Average value: 127 MPa Grade Count:28
Elongation at Break	0.600 - 4.10 %		0.600 - 4.10 %	Average value: 2.07 % Grade Count:28
Modulus of Elasticity	7.10 - 18.0 GPa		1030 - 2610 ksi	Average value: 12.0 GPa Grade Count:15
Flexural Yield Strength	56.0 - 220 MPa		8120 - 31900 psi	Average value: 160 MPa Grade Count:26
Flexural Modulus	7.03 - 20.0 GPa		1020 - 2900 ksi	Average value: 12.2 GPa Grade Count:27

Flexural Strain at Break	1.30 - 4.00 %	1.30 - 4.00 %	Average value: 2.34 % Grade Count:13
Izod Impact, Notched (ISO)	14.0 - 95.0 kJ/m²	6.66 - 45.2 ft-lb/in²	Average value: 68.0 kJ/m² Grade Count:3
Charpy Impact Unnotched	0.336 - 26.7 J/cm²	1.60 - 127 ft-lb/in²	Average value: 5.26 J/cm² Grade Count:10
Charpy Impact, Notched	0.252 - 9.50 J/cm²	1.20 - 45.2 ft-lb/in²	Average value: 1.80 J/cm² Grade Count:20

Electrical Properties	Metric	English	Comments
Electrical Resistivity	1.00e+13 - 1.00e+17 ohm-cm	1.00e+13 - 1.00e+17 ohm-cm	Average value: 1.77e+16 ohm-cm Grade Count:19
Surface Resistance	4.10e+12 - 1.00e+15 ohm	4.10e+12 - 1.00e+15 ohm	Average value: 3.77e+14 ohm Grade Count:6
Dielectric Constant	2.70 - 12.0	2.70 - 12.0	Average value: 4.23 Grade Count:20
Dielectric Strength	19.0 - 53.0 kV/mm	483 - 1350 kV/in	Average value: 32.5 kV/mm Grade Count:14
Dissipation Factor	0.00290 - 0.0300	0.00290 - 0.0300	Average value: 0.0165 Grade Count:16
Arc Resistance	47.0 - 183 sec	47.0 - 183 sec	Average value: 142 sec Grade Count:13
Comparative Tracking Index	100 - 275 V	100 - 275 V	Average value: 166 V Grade Count:16

Thermal Properties	Metric	English	Comments
CTE, linear	6.48 - 34.0 µm/m-°C @Temperature 20.0 - 250 °C	3.60 - 18.9 µin/in-°F @Temperature 68.0 - 482 °F	Average value: 17.5 µm/m-°C Grade Count:3
CTE, linear, Transverse to Flow	32.0 - 64.0 µm/m-°C @Temperature 20.0 - 250 °C	17.8 - 35.6 µin/in-°F @Temperature 68.0 - 482 °F	Average value: 48.4 µm/m-°C Grade Count:3
Thermal Conductivity	0.500 - 34.6 W/m-K	3.47 - 240 BTU-in/hr-ft²-°F	Average value: 5.06 W/m-K Grade Count:5
Melting Point	212 - 350 °C	414 - 662 °F	Average value: 305 °C Grade Count:10
Deflection Temperature at 0.46 MPa (66 psi)	266 - 340 °C	511 - 644 °F	Average value: 291 °C Grade Count:11
Deflection Temperature at 1.8 MPa (264 psi)	105 - 340 °C	221 - 644 °F	Average value: 251 °C Grade Count:29
Flammability, UL94	V-0	V-0	Grade Count:20

Processing Properties	Metric	English	Comments
Processing Temperature	20.0 - 350 °C	68.0 - 662 °F	Average value: 180 °C Grade Count:5

Nozzle Temperature	290 - 382 °C	554 - 720 °F	Average value: 343 °C Grade Count:10
Die Temperature	225 - 295 °C	437 - 563 °F	Average value: 265 °C Grade Count:3
Melt Temperature	185 - 382 °C	365 - 720 °F	Average value: 308 °C Grade Count:16
Mold Temperature	65.6 - 177 °C	150 - 351 °F	Average value: 110 °C Grade Count:13
Drying Temperature	130 - 180 °C	266 - 356 °F	Average value: 152 °C Grade Count:11
Injection Pressure	50.0 - 150 MPa	7250 - 21800 psi	Average value: 104 MPa Grade Count:7

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.